Multiple linear regression

Guibril Ramde

Grading the professor

Many college courses conclude by giving students the opportunity to evaluate the course and the instructor anonymously. However, the use of these student evaluations as an indicator of course quality and teaching effectiveness is often criticized because these measures may reflect the influence of non-teaching related characteristics, such as the physical appearance of the instructor. The article titled, "Beauty in the classroom: instructors' pulchritude and putative pedagogical productivity" by Hamermesh and Parker found that instructors who are viewed to be better looking receive higher instructional ratings.

Here, you will analyze the data from this study in order to learn what goes into a positive professor evaluation.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(GGally)
```

This is the first time we're using the `GGally` package. You will be using the `ggpairs` function from this package later in the lab.

The data

The data were gathered from end of semester student evaluations for a large sample of professors from the University of Texas at Austin. In addition, six students rated the professors' physical appearance. The result is a data frame where each row contains a different course and columns represent variables about the courses and professors. It's called `evals`.

```
glimpse(evals)
```

```
## Rows: 463
## Columns: 23
## $ course_id     <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ prof_id       <int> 1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5,~
## $ score         <dbl> 4.7, 4.1, 3.9, 4.8, 4.6, 4.3, 2.8, 4.1, 3.4, 4.5, 3.8, 4~
## $ rank          <fct> tenure track, tenure track, tenure track, tenure track, ~
```

```
## $ ethnicity     <fct> minority, minority, minority, minority, not minority, no~
## $ gender        <fct> female, female, female, female, male, male, male, male, ~
## $ language      <fct> english, english, english, english, english, english, en~
## $ age           <int> 36, 36, 36, 36, 59, 59, 59, 51, 51, 40, 40, 40, 40, 40, ~
## $ cls_perc_eval <dbl> 55.81395, 68.80000, 60.80000, 62.60163, 85.00000, 87.500~
## $ cls_did_eval  <int> 24, 86, 76, 77, 17, 35, 39, 55, 111, 40, 24, 24, 17, 14,~
## $ cls_students  <int> 43, 125, 125, 123, 20, 40, 44, 55, 195, 46, 27, 25, 20, ~
## $ cls_level     <fct> upper, upper, upper, upper, upper, upper, upper, upper, ~
## $ cls_profs     <fct> single, single, single, single, multiple, multiple, mult~
## $ cls_credits   <fct> multi credit, multi credit, multi credit, multi credit, ~
## $ bty_f1lower   <int> 5, 5, 5, 5, 4, 4, 4, 5, 5, 2, 2, 2, 2, 2, 2, 2, 2, 7, 7,~
## $ bty_f1upper   <int> 7, 7, 7, 7, 4, 4, 4, 2, 2, 5, 5, 5, 5, 5, 5, 5, 5, 9, 9,~
## $ bty_f2upper   <int> 6, 6, 6, 6, 2, 2, 2, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4, 9, 9,~
## $ bty_m1lower   <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 7, 7,~
## $ bty_m1upper   <int> 4, 4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 6, 6,~
## $ bty_m2upper   <int> 6, 6, 6, 6, 3, 3, 3, 3, 3, 2, 2, 2, 2, 2, 2, 2, 2, 6, 6,~
## $ bty_avg       <dbl> 5.000, 5.000, 5.000, 5.000, 3.000, 3.000, 3.000, 3.333, ~
## $ pic_outfit    <fct> not formal, not formal, not formal, not formal, not form~
## $ pic_color     <fct> color, color, color, color, color, color, color, color, ~
```

We have observations on 21 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file:

```
?evals
```

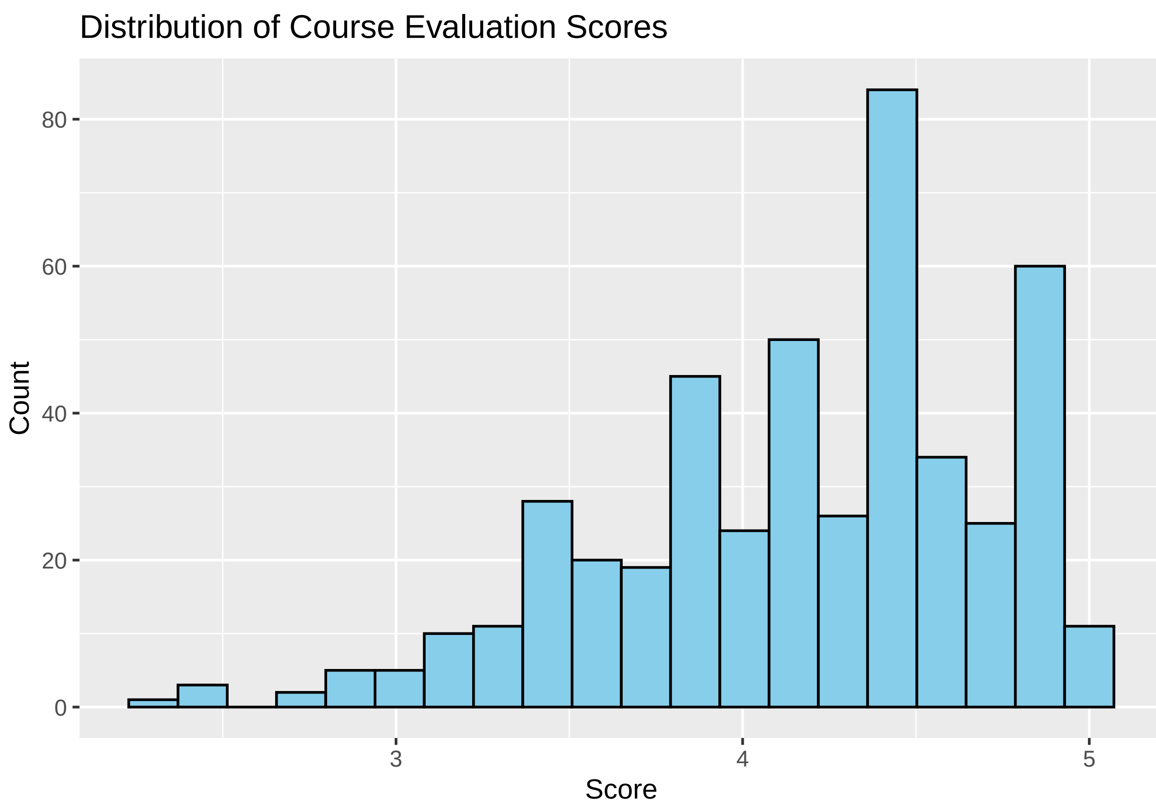
Exploring the data

1. Is this an observational study or an experiment? The original research question posed in the paper is whether beauty leads directly to the differences in course evaluations. Given the study design, is it possible to answer this question as it is phrased? If not, rephrase the question.

Insert your answer here: this is an observational study, no it is not posible to answer the question as it is phrased a better question would be : "Is there an association between professor beauty and course evaluation scores?"

2. Describe the distribution of `score`. Is the distribution skewed? What does that tell you about how students rate courses? Is this what you expected to see? Why, or why not?

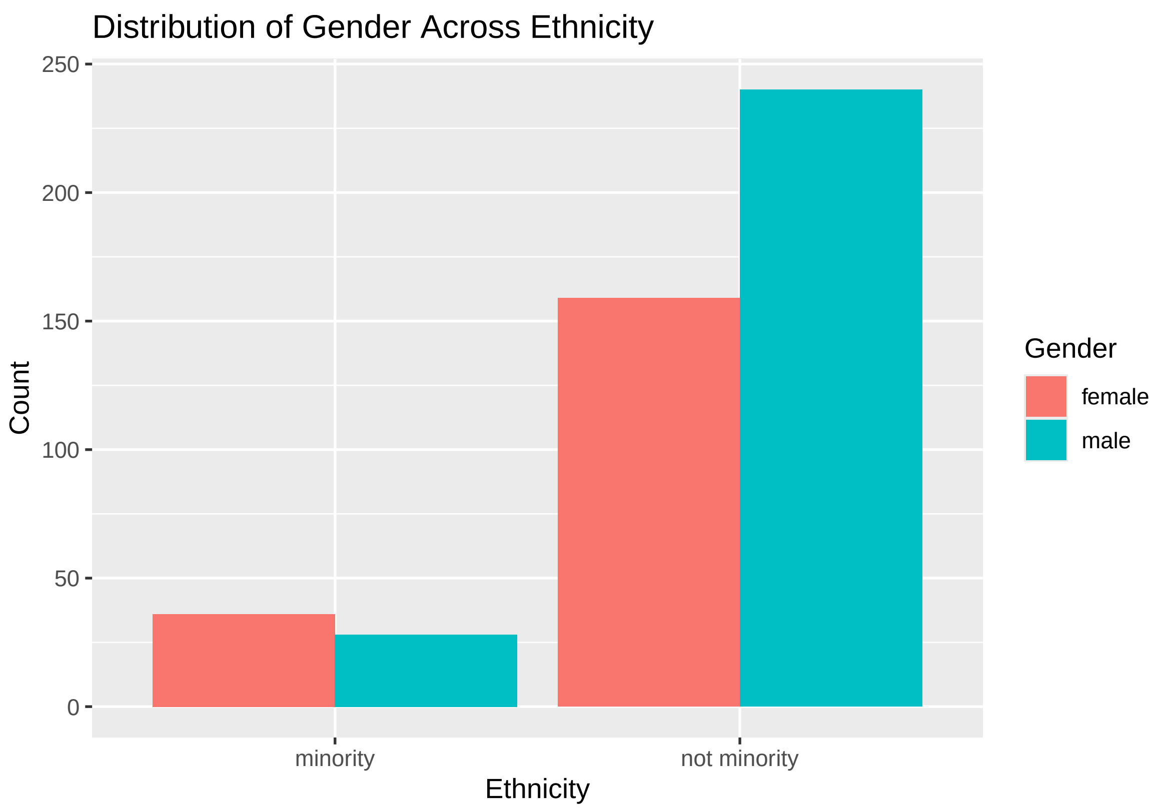
```
ggplot(evals, aes(x = score)) +
  geom_histogram(bins = 20, fill = "skyblue", color = "black") +
  labs(
title = "Distribution of Course Evaluation Scores",
x = "Score",
y = "Count"
  )
```



Insert your answer here:The distribution of score is left-skewed (negatively skewed), meaning most values are concentrated on the higher end (closer to 4–5), with fewer low scores. This indicates that students generally give high ratings to professors. This is expected because course evaluations are often positively biased, with students tending to rate instructors favorably unless they had a very negative experience.

3. Excluding `score`, select two other variables and describe their relationship with each other using an appropriate visualization.

```
   ggplot(evals, aes(x = ethnicity, fill = gender)) +
  geom_bar(position = "dodge") +
  labs(
title = "Distribution of Gender Across Ethnicity",
x = "Ethnicity",
y = "Count",
fill = "Gender"
  )
```

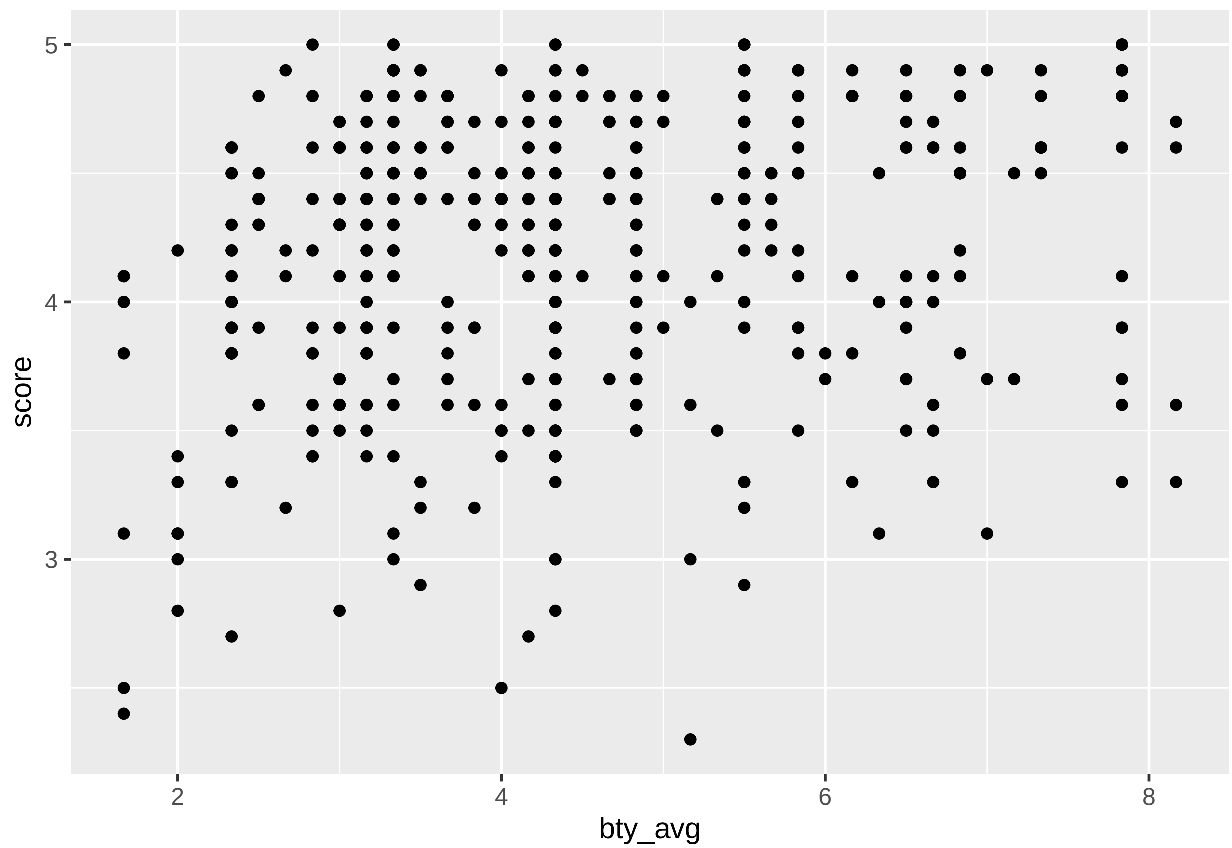


Insert your answer here: The bar chart shows the relationship between ethnicity and gender. Both minority and non-minority groups include male and female instructors, but the counts differ slightly. Overall, there appear to be more non-minority instructors than minority instructors in the dataset. Within each ethnicity group, the distribution of gender is relatively similar, though one gender may slightly dominate depending on the group. This suggests that while both variables vary, there is no strong imbalance between gender within each ethnicity category.

Simple linear regression

The fundamental phenomenon suggested by the study is that better looking teachers are evaluated more favorably. Let's create a scatterplot to see if this appears to be the case:

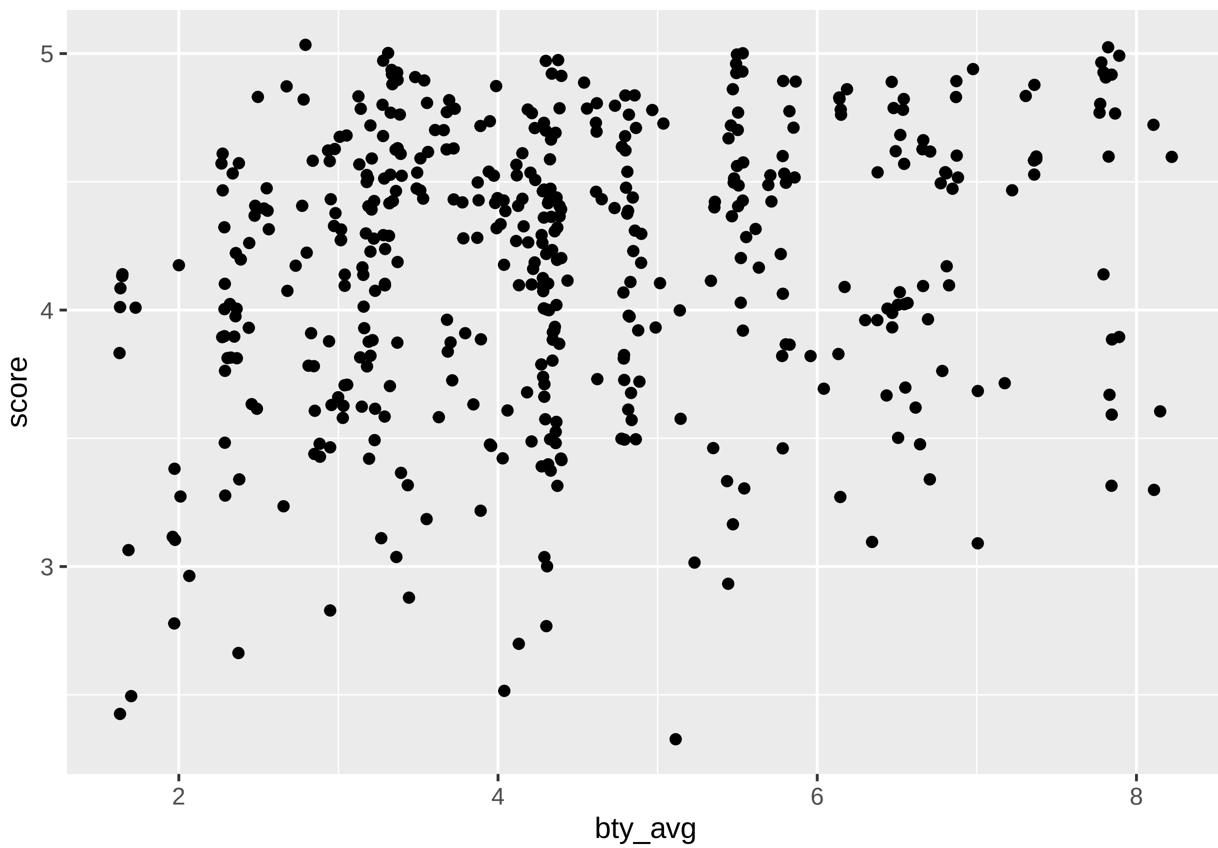
```
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_point()
```



Before you draw conclusions about the trend, compare the number of observations in the data frame with the approximate number of points on the scatterplot. Is anything awry?

4. Replot the scatterplot, but this time use `geom_jitter` as your layer. What was misleading about the initial scatterplot?

```
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter()
```



Insert your answer here: the initial scatterplot was scatter and show less observation but using geom_jitter we get a good representation of the observation

5. Let's see if the apparent trend in the plot is something more than natural variation. Fit a linear model called `m_bty` to predict average professor score by average beauty rating. Write out the equation for the linear model and interpret the slope. Is average beauty score a statistically significant predictor? Does it appear to be a practically significant predictor?

```
m_bty <- lm(score ~ bty_avg, data = evals)
m_bty
```

```
##
## Call:
## lm(formula = score ~ bty_avg, data = evals)
##
## Coefficients:
## (Intercept)      bty_avg
##     3.88034      0.06664
```

```
anova(m_bty)
```

```
## Analysis of Variance Table
##
## Response: score
##            Df  Sum Sq Mean Sq F value    Pr(>F)
```

```
## bty_avg     1   4.786  4.7859  16.731 5.083e-05 ***
## Residuals 461 131.868  0.2860
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_bty)
```

```
##
## Call:
## lm(formula = score ~ bty_avg, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9246 -0.3690  0.1420  0.3977  0.9309
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.88034    0.07614   50.96  < 2e-16 ***
## bty_avg      0.06664    0.01629    4.09 5.08e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5348 on 461 degrees of freedom
## Multiple R-squared:  0.03502,    Adjusted R-squared:  0.03293
## F-statistic: 16.73 on 1 and 461 DF,  p-value: 5.083e-05
```

```r
predictions <- predict(m_bty)
```

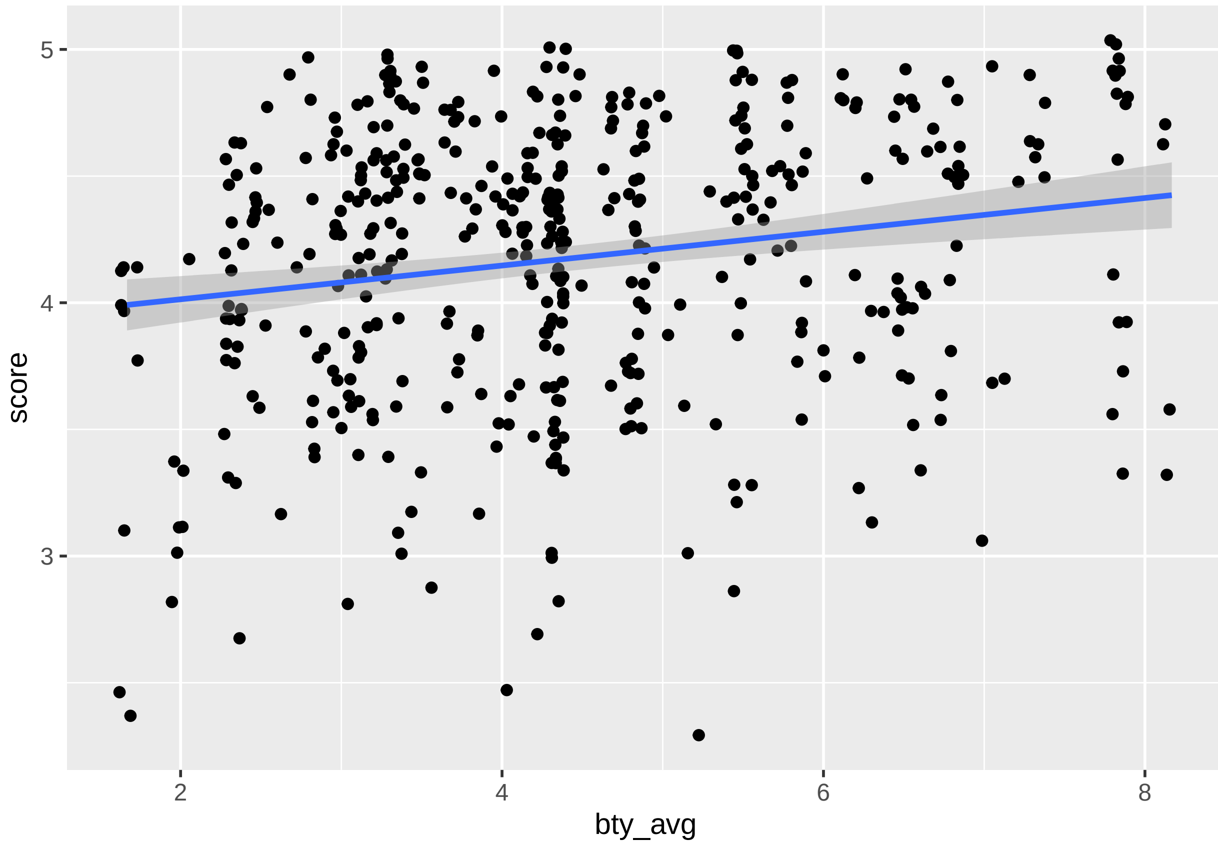
Insert your answer here:

$$\hat{score} = 3.88 + 0.066 \times bty_avg$$

Answer: The slope of 0.0666 indicates that for each one-unit increase in average beauty rating, the predicted course evaluation score increases by about 0.067 points on average. The p-value for bty_avg is very small (p = 5.08e-05), which is less than 0.05. This indicates that average beauty score is a statistically significant predictor of course evaluation score. However, the R square value is only 0.035, meaning that beauty explains about 3.5% of the variability in course evaluation scores. This is a very small proportion, and the slope is also quite small. Therefore, while the relationship is statistically significant, it is not practically significant, since changes in beauty rating lead to only minimal increases in evaluation scores.

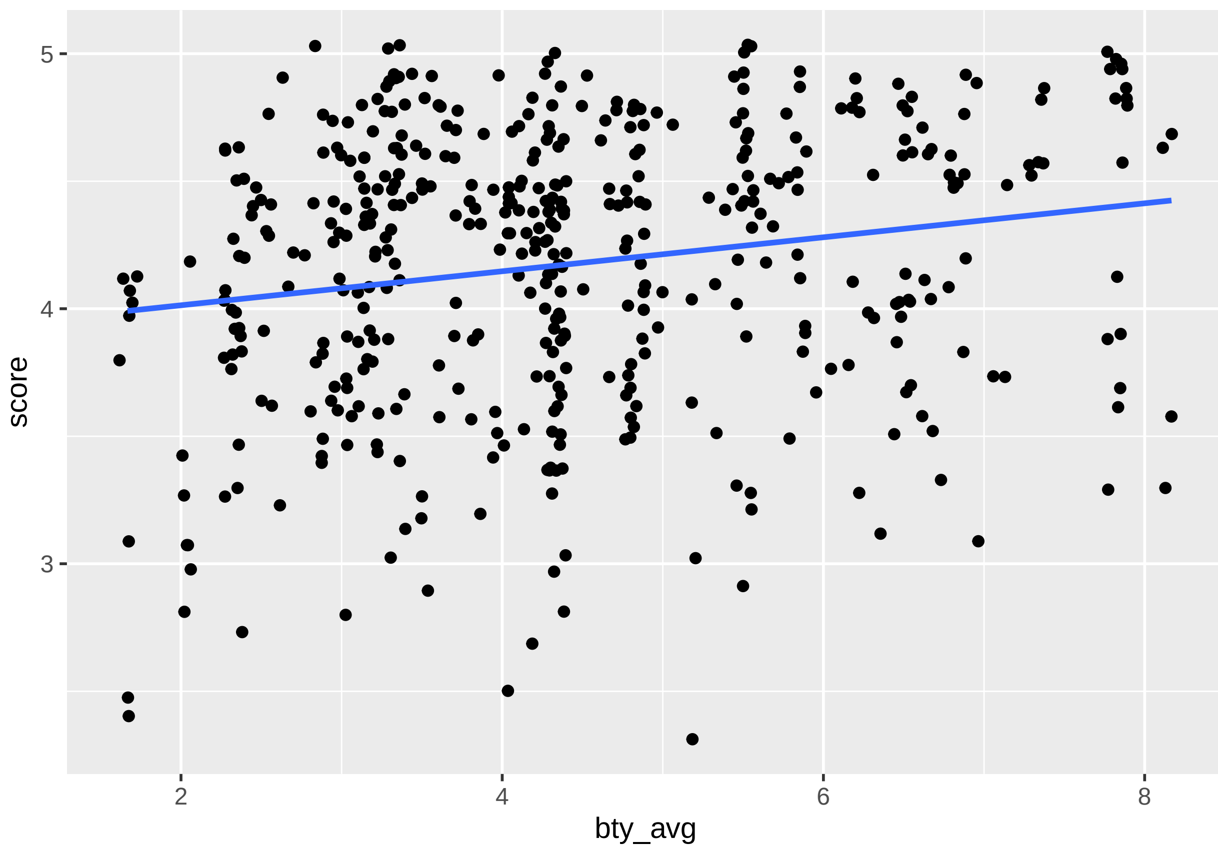
Add the line of the best fit model to your plot using the following:

```r
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter() +
  geom_smooth(method = "lm")
```



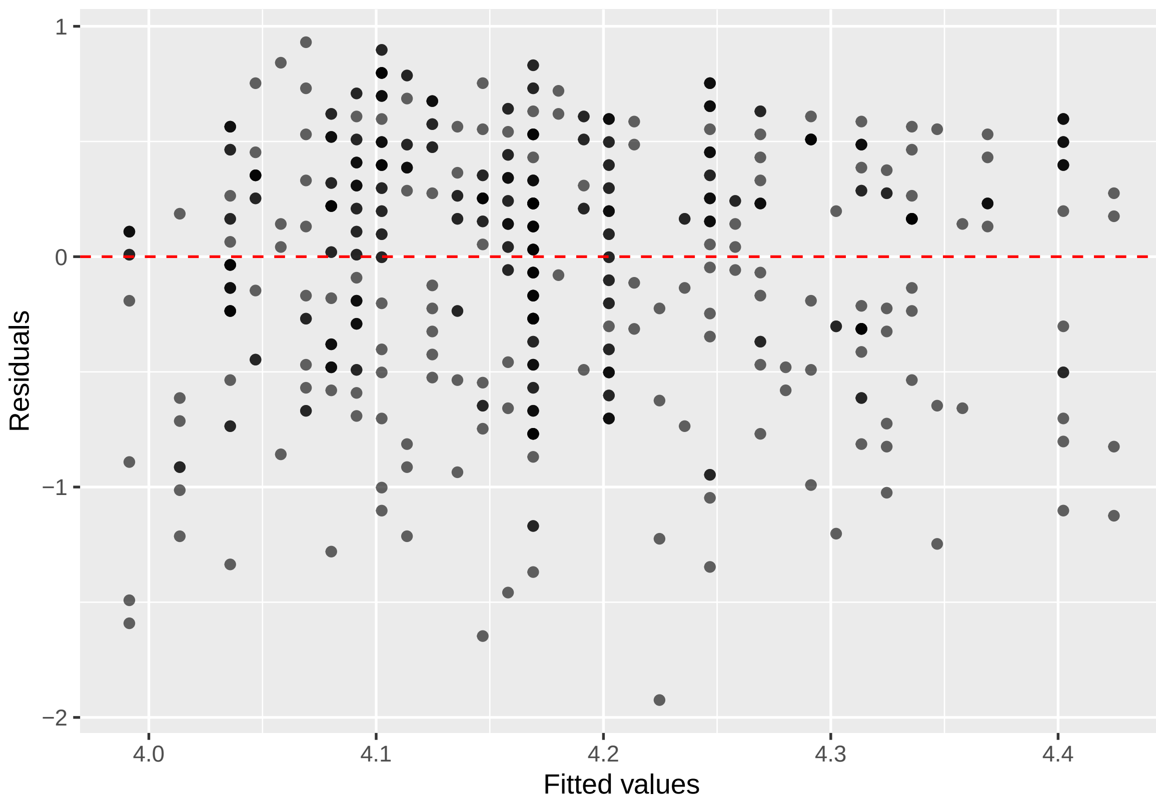
The blue line is the model. The shaded gray area around the line tells you about the variability you might expect in your predictions. To turn that off, use `se = FALSE`.

```
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter() +
  geom_smooth(method = "lm", se = FALSE)
```

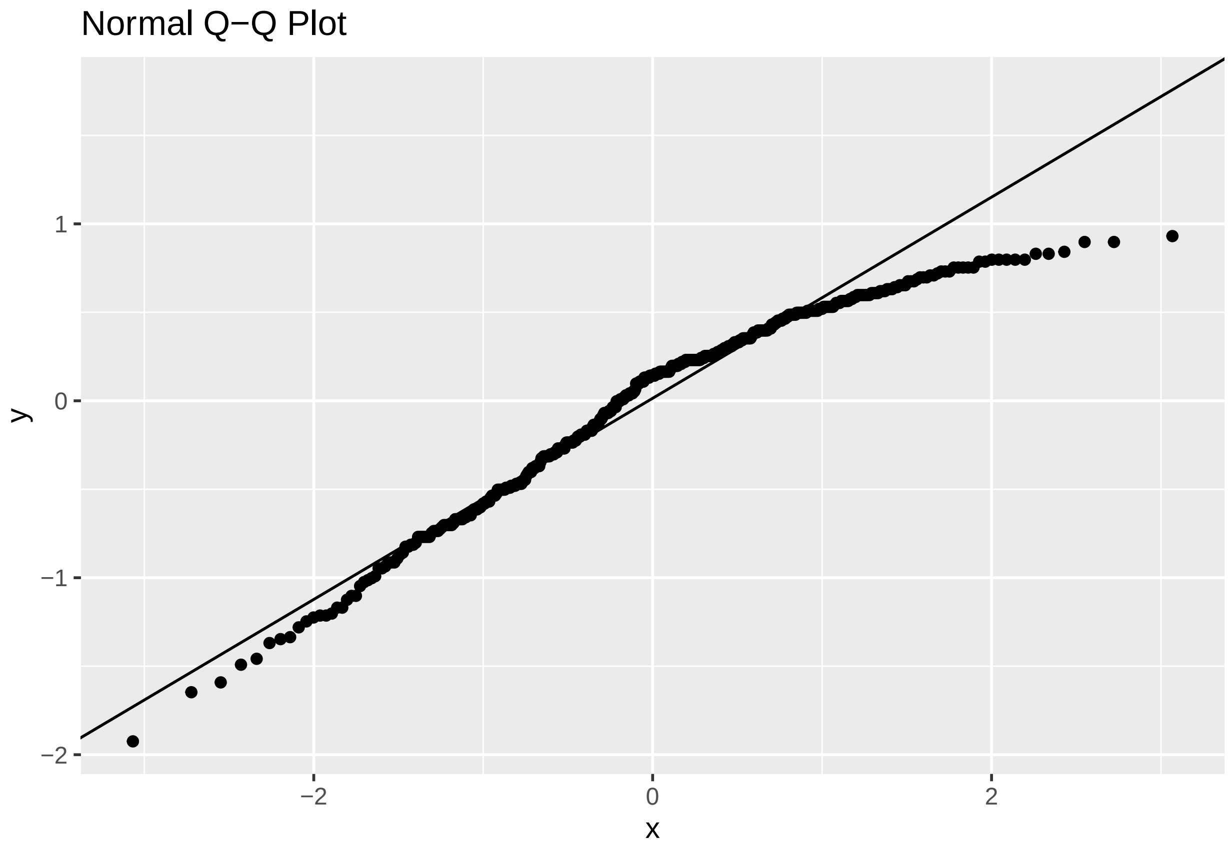


6. Use residual plots to evaluate whether the conditions of least squares regression are reasonable. Provide plots and comments for each one (see the Simple Regression Lab for a reminder of how to make these).

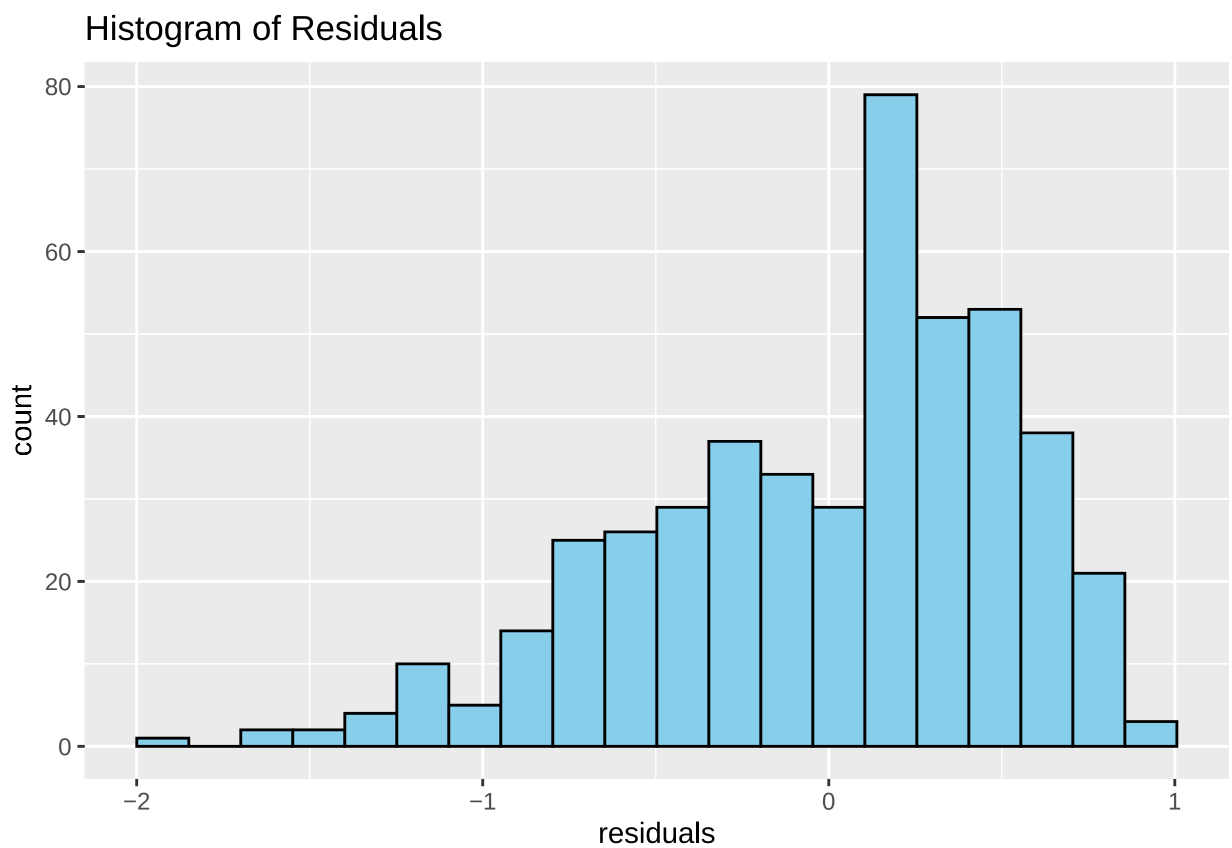
```
   #evals$residuals <- resid(m_bty)
#evals$fitted <- fitted(m_bty)
#ggplot(evals, aes(x = fitted, y = residuals)) +
#   geom_point(alpha = 0.6) +
 # geom_hline(yintercept = 0, color = "red") +
 # labs(
 #   title = "Residuals vs Fitted Values",
  #  x = "Fitted Values",
  #  y = "Residuals"
 # )

ggplot(data = m_bty, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  xlab("Fitted values") +
  ylab("Residuals")
```




```
evals$residuals <- resid(m_bty)
ggplot(evals, aes(sample = residuals)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Normal Q-Q Plot")
```




```r
ggplot(evals, aes(x = residuals)) +
  geom_histogram(bins = 20, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Residuals")
```

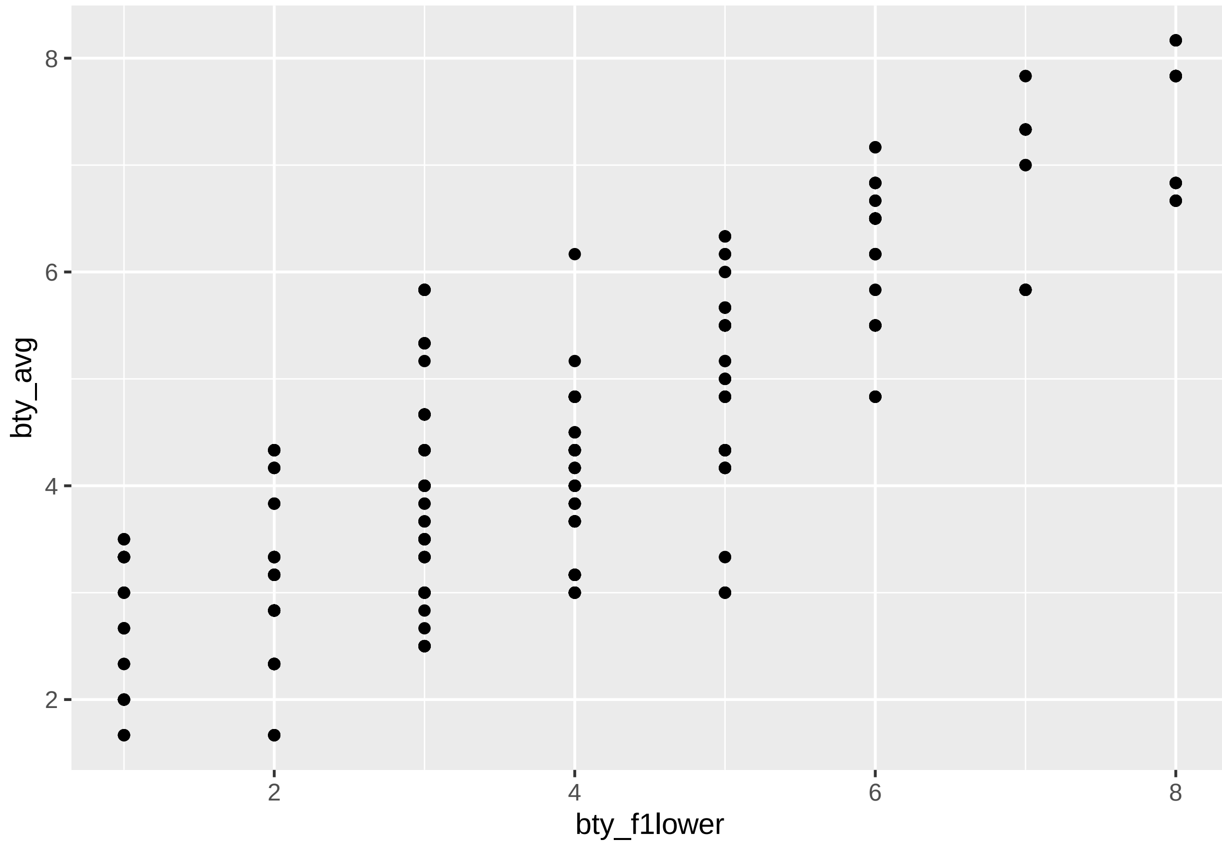


Insert your answer here: The residuals vs fitted plot shows that the residuals are randomly scattered around zero with no clear pattern, suggesting that the linearity assumption is reasonable. There is no strong evidence of non-linearity. The spread of residuals appears to be relatively constant across fitted values, indicating that the constant variance (homoscedasticity) assumption is reasonably satisfied.The Q-Q plot shows that the residuals fall approximately along the straight line, suggesting that the normality assumption is also reasonable, although there may be slight deviations at the tails. Overall, the conditions for least squares regression appear to be reasonably met

Multiple linear regression

The data set contains several variables on the beauty score of the professor: individual ratings from each of the six students who were asked to score the physical appearance of the professors and the average of these six scores. Let's take a look at the relationship between one of these scores and the average beauty score.

```
ggplot(data = evals, aes(x = bty_f1lower, y = bty_avg)) +
  geom_point()
```

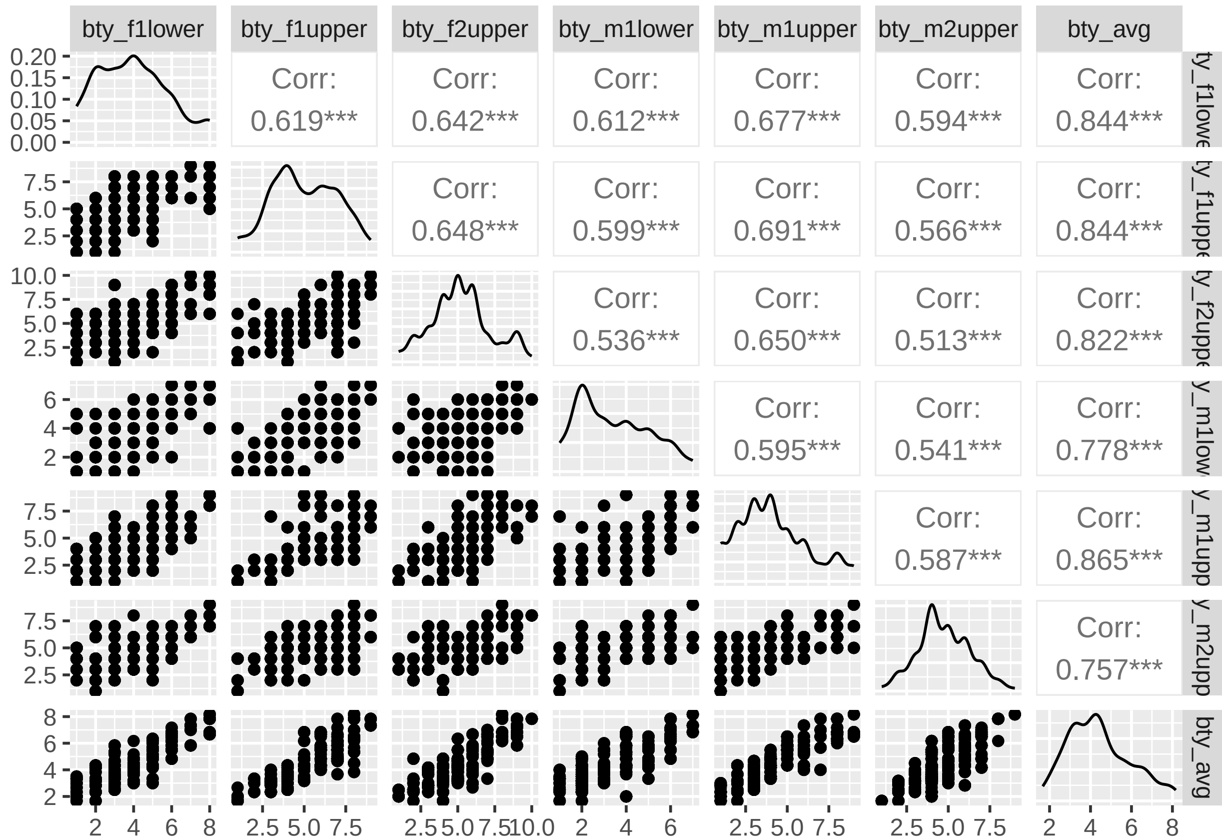



```
evals %>%
  summarise(cor(bty_avg, bty_f1lower))
```

```
## # A tibble: 1 x 1
##   `cor(bty_avg, bty_f1lower)`
##                         <dbl>
## 1                       0.844
```

As expected, the relationship is quite strong—after all, the average score is calculated using the individual scores. You can actually look at the relationships between all beauty variables (columns 13 through 19) using the following command:

```
#install.packages("GGally")
library(GGally)
evals %>%
  select(contains("bty")) %>%
  ggpairs()
```



These variables are collinear (correlated), and adding more than one of these variables to the model would not add much value to the model. In this application and with these highly-correlated predictors, it is reasonable to use the average beauty score as the single representative of these variables.

In order to see if beauty is still a significant predictor of professor score after you've accounted for the professor's gender, you can add the gender term into the model.

```
m_bty_gen <- lm(score ~ bty_avg + gender, data = evals)
summary(m_bty_gen)
```

```
##
## Call:
## lm(formula = score ~ bty_avg + gender, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8305 -0.3625  0.1055  0.4213  0.9314
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.74734    0.08466  44.266  < 2e-16 ***
## bty_avg      0.07416    0.01625   4.563 6.48e-06 ***
## gendermale   0.17239    0.05022   3.433 0.000652 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.5287 on 460 degrees of freedom
## Multiple R-squared:  0.05912,    Adjusted R-squared:  0.05503
## F-statistic: 14.45 on 2 and 460 DF,  p-value: 8.177e-07
```

7. P-values and parameter estimates should only be trusted if the conditions for the regression are reasonable. Verify that the conditions for this model are reasonable using diagnostic plots.

Insert your answer here:The diagnostic plots indicate that the conditions for linear regression are reasonably satisfied. The residuals appear randomly scattered around zero, suggesting linearity. The spread of residuals is relatively constant, indicating homoscedasticity. The Q-Q plot shows approximate normality, with only slight deviations at the tails. Therefore, the model assumptions appear to be reasonable..

8. Is `bty_avg` still a significant predictor of `score`? Has the addition of `gender` to the model changed the parameter estimate for `bty_avg`?

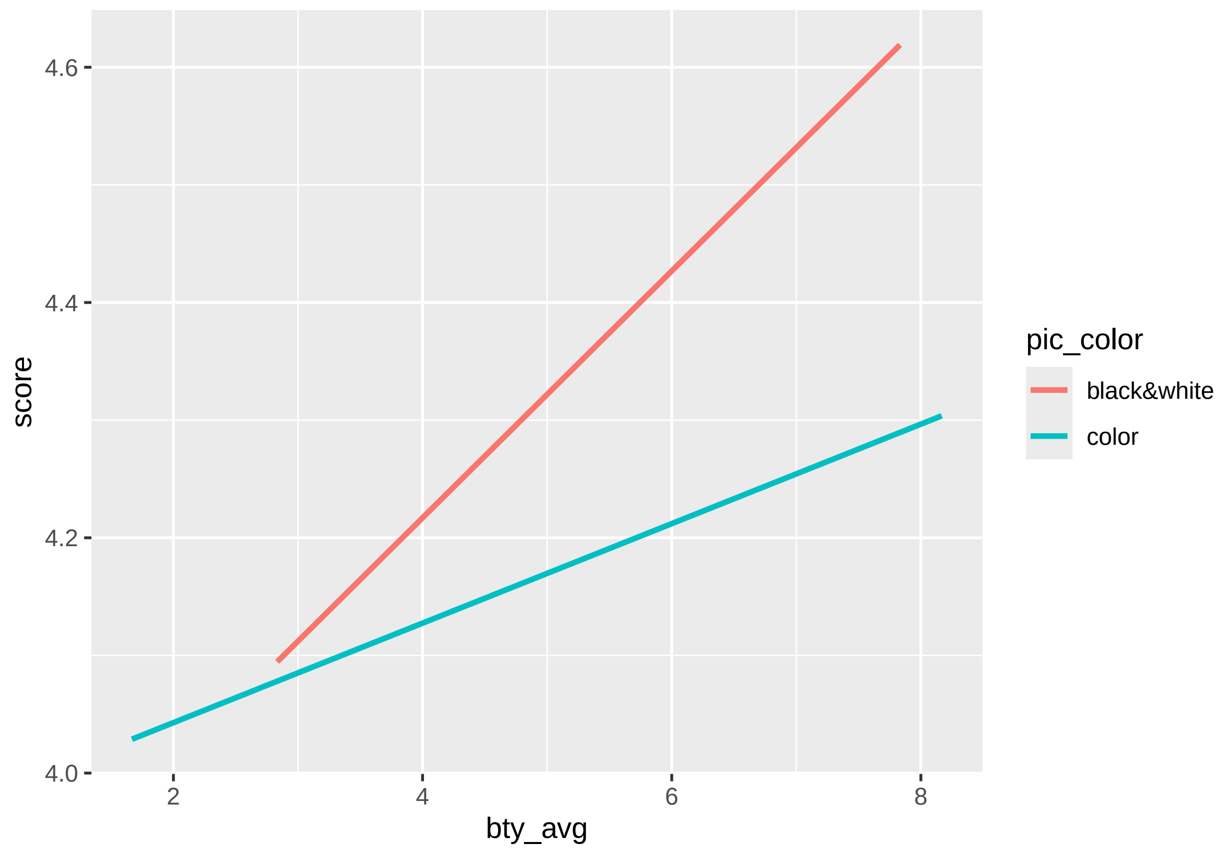
Insert your answer here: Yes, bty__avg remains a statistically significant predictor of score (p-value = 6.48e-06) even after including gender in the model. The coefficient for bty__avg increased slightly from about 0.0666 to 0.0742, indicating a slightly stronger relationship after accounting for gender. This suggests that beauty continues to have an independent effect on course evaluations. Additionally, the coefficient for gender indicates that male professors receive, on average, about 0.17 higher evaluation scores than female professors, holding beauty constant.

Note that the estimate for `gender` is now called `gendermale`. You'll see this name change whenever you introduce a categorical variable. The reason is that R recodes `gender` from having the values of `male` and `female` to being an indicator variable called `gendermale` that takes a value of 0 for female professors and a value of 1 for male professors. (Such variables are often referred to as "dummy" variables.)

As a result, for female professors, the parameter estimate is multiplied by zero, leaving the intercept and slope form familiar from simple regression.

$$
\begin{aligned}
\widehat{score} &= \hat{\beta}_0 + \hat{\beta}_1 \times bty\_avg + \hat{\beta}_2 \times (0) \\
&= \hat{\beta}_0 + \hat{\beta}_1 \times bty\_avg
\end{aligned}
$$

```r
ggplot(data = evals, aes(x = bty_avg, y = score, color = pic_color)) +
 geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```



9. What is the equation of the line corresponding to those with color pictures? (*Hint:* For those with color pictures, the parameter estimate is multiplied by 1.) For two professors who received the same beauty rating, which color picture tends to have the higher course evaluation score?

**Insert your answer here:The equation for professors with color pictures is:*

$$\hat{score} = \beta_0 + \beta_1 \cdot bty_avg + \beta_2$$

Answer: Since the indicator variable equals 1 for color pictures, the coefficient β_2 shifts the regression line upward or downward. If β_2 is positive,professors with color pictures tend to have higher evaluation scores; if negative, those with black and white pictures tend to have higher scores.

The decision to call the indicator variable `gendermale` instead of `genderfemale` has no deeper meaning. R simply codes the category that comes first alphabetically as a 0. (You can change the reference level of a categorical variable, which is the level that is coded as a 0, using the`relevel()` function. Use `?relevel` to learn more.)

10. Create a new model called `m_bty_rank` with `gender` removed and `rank` added in. How does R appear to handle categorical variables that have more than two levels? Note that the rank variable has three levels: `teaching`, `tenure track`, `tenured`.

Insert your answer here: R handles categorical variables with more than two levels by creating multiple dummy variables. Since rank has three levels, R creates two indicator variables, using one category (teaching) as the reference group. The coefficients for the other levels represent how much their scores differ from the reference category

The interpretation of the coefficients in multiple regression is slightly different from that of simple regression. The estimate for `bty_avg` reflects how much higher a group of professors is expected to score if they have a beauty rating that is one point higher *while holding all other variables constant.* In this case, that translates into considering only professors of the same rank with `bty_avg` scores that are one point apart.

The search for the best model

We will start with a full model that predicts professor score based on rank, gender, ethnicity, language of the university where they got their degree, age, proportion of students that filled out evaluations, class size, course level, number of professors, number of credits, average beauty rating, outfit, and picture color.

11. Which variable would you expect to have the highest p-value in this model? Why? *Hint:* Think about which variable would you expect to not have any association with the professor score.

Insert your answer here:I would expect variables such as cls_profssingle or pic_outfit to have the highest p-values, since they are less likely to have a meaningful relationship with course evaluation scores. These variables do not directly reflect teaching quality or student experience, so they are less likely to be strong predictors.

Let's run the model...

```
m_full <- lm(score ~ rank + gender + ethnicity + language + age + cls_perc_eval
             + cls_students + cls_level + cls_profs + cls_credits + bty_avg
             + pic_outfit + pic_color, data = evals)
summary(m_full)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
##     cls_perc_eval + cls_students + cls_level + cls_profs + cls_credits +
##     bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.77397 -0.32432  0.09067  0.35183  0.95036
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.0952141  0.2905277  14.096  < 2e-16 ***
## ranktenure track      -0.1475932  0.0820671  -1.798  0.07278 .
## ranktenured           -0.0973378  0.0663296  -1.467  0.14295
## gendermale             0.2109481  0.0518230   4.071 5.54e-05 ***
## ethnicitynot minority  0.1234929  0.0786273   1.571  0.11698
## languagenon-english   -0.2298112  0.1113754  -2.063  0.03965 *
## age                   -0.0090072  0.0031359  -2.872  0.00427 **
## cls_perc_eval          0.0053272  0.0015393   3.461  0.00059 ***
## cls_students           0.0004546  0.0003774   1.205  0.22896
## cls_levelupper         0.0605140  0.0575617   1.051  0.29369
## cls_profssingle       -0.0146619  0.0519885  -0.282  0.77806
## cls_creditsone credit  0.5020432  0.1159388   4.330 1.84e-05 ***
## bty_avg                0.0400333  0.0175064   2.287  0.02267 *
## pic_outfitnot formal  -0.1126817  0.0738800  -1.525  0.12792
## pic_colorcolor        -0.2172630  0.0715021  -3.039  0.00252 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.498 on 448 degrees of freedom
## Multiple R-squared:  0.1871, Adjusted R-squared:  0.1617
## F-statistic: 7.366 on 14 and 448 DF,  p-value: 6.552e-14
```

12. Check your suspicions from the previous exercise. Include the model output in your response.

Insert your answer here: The variable with the highest p-value is cls_profssingle (p = 0.778), confirming the expectation that this variable is not a meaningful predictor of course evaluation scores. Other variables with high p-values include cls_levelupper and cls_students, which also do not appear to be statistically significant predictors in this model

13. Interpret the coefficient associated with the ethnicity variable.

Insert your answer here: The coefficient for ethnicity represents the difference in evaluation scores between minority and non-minority professors, holding all other variables constant. A positive coefficient would indicate higher scores for minority professors, while a negative coefficient would indicate lower scores compared to non-minority professors.

14. Drop the variable with the highest p-value and re-fit the model. Did the coefficients and significance of the other explanatory variables change? (One of the things that makes multiple regression interesting is that coefficient estimates depend on the other variables that are included in the model.) If not, what does this say about whether or not the dropped variable was collinear with the other explanatory variables?

```r
m_reduced1 <- lm(score ~ rank   + gender  + ethnicity + language + age +
             cls_perc_eval + cls_students + cls_level +
             cls_credits + bty_avg + pic_outfit + pic_color,
             data = evals)

summary(m_reduced1)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
##     cls_perc_eval + cls_students + cls_level + cls_credits +
##     bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7836 -0.3257  0.0859  0.3513  0.9551
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.0872523  0.2888562  14.150  < 2e-16 ***
## ranktenure track      -0.1476746  0.0819824  -1.801 0.072327 .
## ranktenured           -0.0973829  0.0662614  -1.470 0.142349
## gendermale             0.2101231  0.0516873   4.065 5.66e-05 ***
## ethnicitynot minority  0.1274458  0.0772887   1.649 0.099856 .
## languagenon-english   -0.2282894  0.1111305  -2.054 0.040530 *
```

```
## age                    -0.0089992  0.0031326  -2.873 0.004262 **
## cls_perc_eval           0.0052888  0.0015317   3.453 0.000607 ***
## cls_students            0.0004687  0.0003737   1.254 0.210384
## cls_levelupper          0.0606374  0.0575010   1.055 0.292200
## cls_creditsone credit  0.5061196  0.1149163   4.404 1.33e-05 ***
## bty_avg                 0.0398629  0.0174780   2.281 0.023032 *
## pic_outfitnot formal   -0.1083227  0.0721711  -1.501 0.134080
## pic_colorcolor         -0.2190527  0.0711469  -3.079 0.002205 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4974 on 449 degrees of freedom
## Multiple R-squared:  0.187,  Adjusted R-squared:  0.1634
## F-statistic: 7.943 on 13 and 449 DF,  p-value: 2.336e-14
```

Insert your answer here:The variable cls_profssingle, which had the highest p-value, was removed and the model was refitted. After removing this variable, the coefficients and p-values of the remaining predictors did not change substantially. Additionally, the R square value remained nearly the same. This suggests that cls_profssingle was not an important predictor and did not contribute meaningful information to the model. It also indicates that this variable was not strongly collinear with the other explanatory variables.

15. Using backward-selection and p-value as the selection criterion, determine the best model. You do not need to show all steps in your answer, just the output for the final model. Also, write out the linear model for predicting score based on the final model you settle on.

**Insert your answer here: Using backward selection, variables with the highest p-values were removed one at a time until only statistically significant predictors remained. The final model includes predictors such as gender, language, age, cls_perc_eval, cls_credits, bty_avg, and pic_color.*

The final model can be written as:

```
m_final <- lm(score ~ gender + language + age + cls_perc_eval +
              cls_credits + bty_avg + pic_color,
              data = evals)

summary(m_final)
```

```
##
## Call:
## lm(formula = score ~ gender + language + age + cls_perc_eval +
##     cls_credits + bty_avg + pic_color, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.81919 -0.32035  0.09272  0.38526  0.88213
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              3.967255   0.215824  18.382  < 2e-16 ***
## gendermale               0.221457   0.049937   4.435 1.16e-05 ***
## languagenon-english     -0.281933   0.098341  -2.867  0.00434 **
## age                     -0.005877   0.002622  -2.241  0.02551 *
```

```
## cls_perc_eval          0.004295   0.001432   2.999  0.00286 **
## cls_creditsone credit  0.444392   0.100910   4.404 1.33e-05 ***
## bty_avg                0.048679   0.016974   2.868  0.00432 **
## pic_colorcolor        -0.216556   0.066625  -3.250  0.00124 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5014 on 455 degrees of freedom
## Multiple R-squared:  0.1631, Adjusted R-squared:  0.1502
## F-statistic: 12.67 on 7 and 455 DF,  p-value: 6.996e-15
```

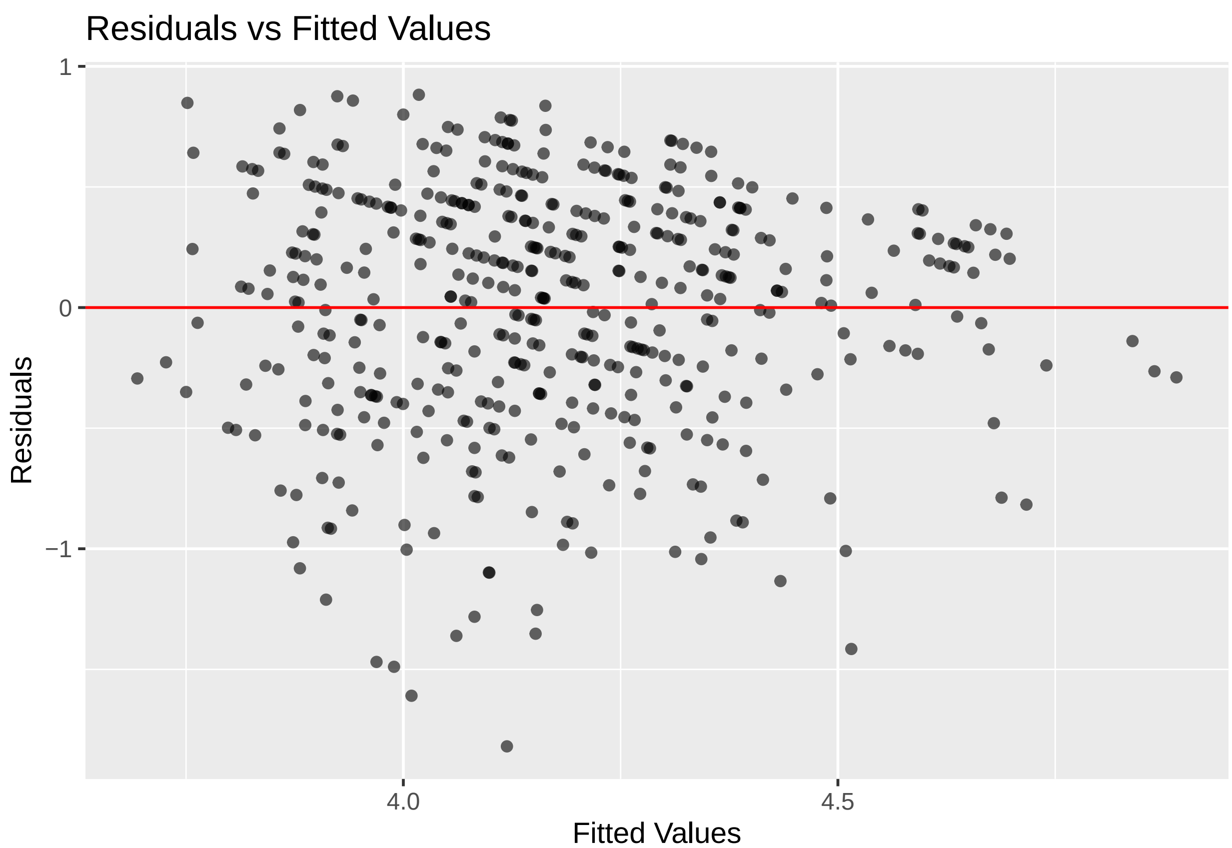
```r
#predictions <- predict(m_final)
#predictions
```

$$\hat{score} = 3.967 + 0.221 \cdot gender\_male - 0.282 \cdot languagenon\_english - 0.0059 \cdot age + 0.0043 \cdot cls\_perc\_eval + 0.444 \cdot cls\_creditsone\_$$

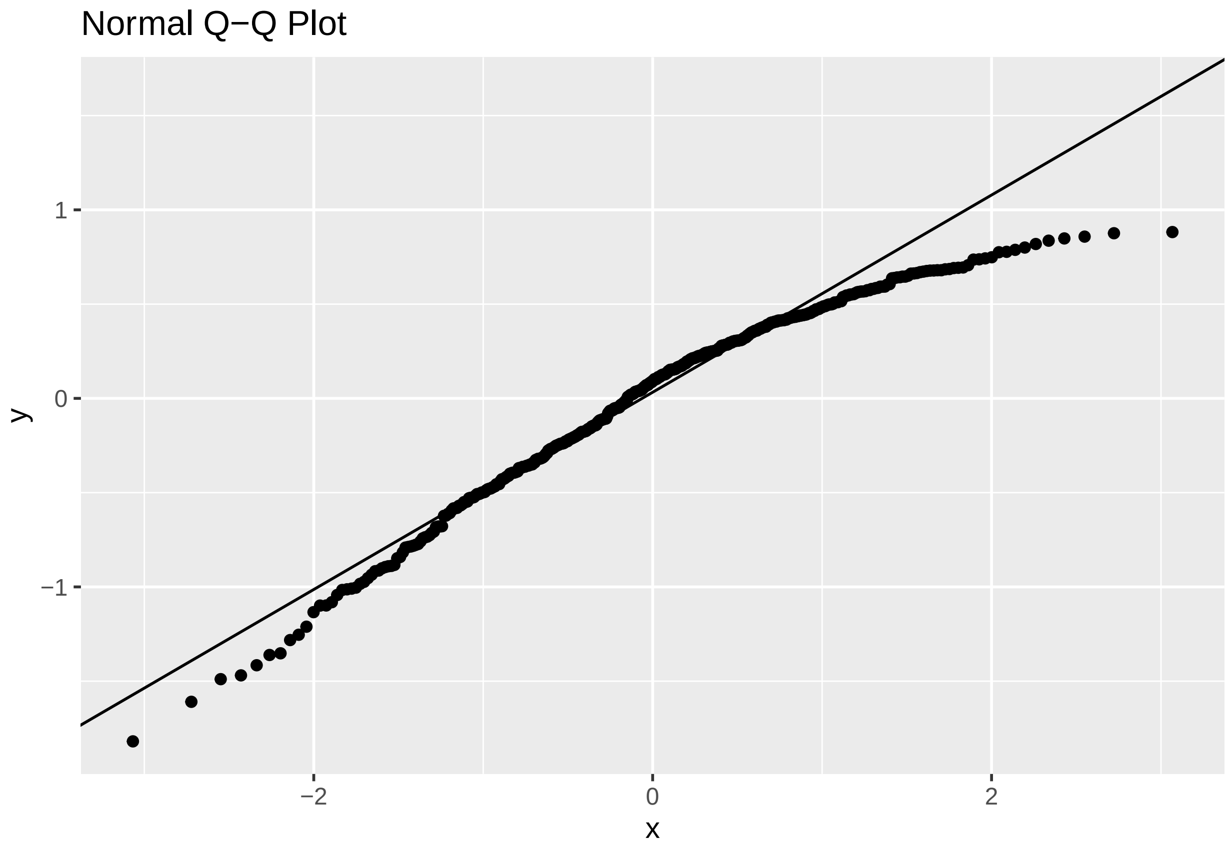
16. Verify that the conditions for this model are reasonable using diagnostic plots.

```r
# get residuals and fitted values
evals$residuals_final <- resid(m_final)
evals$fitted_final <- fitted(m_final)
```

```r
ggplot(evals, aes(x = fitted_final, y = residuals_final)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, color = "red") +
  labs(
    title = "Residuals vs Fitted Values",
    x = "Fitted Values",
    y = "Residuals"
  )
```




```r
ggplot(evals, aes(sample = residuals_final)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Normal Q-Q Plot")
```



Insert your answer here: The diagnostic plots for the final model indicate that the conditions for linear regression are reasonably satisfied. The residuals vs fitted plot shows no clear pattern, suggesting that the linearity assumption holds. The spread of residuals appears relatively constant across fitted values, indicating homoscedasticity. The Q-Q plot shows that the residuals are approximately normally distributed, with only slight deviations at the tails. Overall, the model assumptions appear to be reasonable.

17. The original paper describes how these data were gathered by taking a sample of professors from the University of Texas at Austin and including all courses that they have taught. Considering that each row represents a course, could this new information have an impact on any of the conditions of linear regression?

Insert your answer here: Yes, this could impact the independence assumption of linear regression. Since multiple courses may be taught by the same professor, observations are not completely independent. This violates the independence condition and may affect the validity of the model.

18. Based on your final model, describe the characteristics of a professor and course at University of Texas at Austin that would be associated with a high evaluation score.

Insert your answer here: Based on the final model, higher evaluation scores are associated with higher beauty ratings, male gender, higher class evaluation participation percentage, and one-credit courses. Lower evaluation scores are associated with non-English language background, older age, and color pictures. Therefore, a professor/course associated with a high evaluation score would likely have a higher beauty rating, be male, have high student evaluation participation, teach a one-credit course, and not have the negative factors shown in the model.

19. Would you be comfortable generalizing your conclusions to apply to professors generally (at any university)? Why or why not?

Insert your answer here: No, the results cannot be fully generalized to all professors. The data come from a single university, so the findings may not apply to other institutions with different student populations, evaluation systems, or cultural factors

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